# Heet Mehta

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## TECHNICAL SKILLS

- **Technical:** Python, R, SQL, HTML, CSS, C++, MATLAB, kdb+/q, MS Excel, AWS, Git, Bloomberg Terminal, SPSS, VBA, Databricks
- **Financial:** Derivatives Pricing (Black-Scholes Model), Value at Risk (VaR) analysis (Monte Carlo Simulation), Backtesting, Object Oriented Programming (OOP), Linear Regression, Random Forest, Logistic Regression

## EDUCATION

**FORDHAM UNIVERSITY, GABELLI SCHOOL OF BUSINESS** New York, NY
**MS, Quantitative Finance (STEM), Expected May 2027** Aug 2025 - May 2027

- Relevant Coursework: Derivatives, Financial Markets and Modeling, Financial Programming, Machine Learning, Introduction to Stochastic Calculus, Risk Management, Time Series Econometrics, Fixed Income Securities
- Graduate Research Assistant: Data analysis for 'fear of appearing incompetent construct'; Statistical Analysis and Predictive Modeling
- Graduate Research Assistant: Built Python pipeline for 87 million business records; Advanced data collection and engineering

**CALIFORNIA STATE UNIVERSITY, SAN BERNARDINO** Ahmedabad, Gujarat, India
**AHMEDABAD MANAGEMENT ASSOCIATION (Joint Degree in Financial Management), GPA: 4.0/4.0** July 2024 - May 2025

- Relevant Coursework: Capital Budgeting, Mergers and Acquisitions (M&A), Financial Statement Analysis, Foreign Exchange

**INDUS INSTITUTE OF TECHNOLOGY AND ENGINEERING** Ahmedabad, Gujarat, India
**BTech, Computer Science and Engineering, GPA: 3.6/4.0** Sept 2020 - Apr 2024

- Relevant Coursework: Data Science, Data Preparation and Analysis, Cloud Computing, Design and Analysis of Algorithms, Data Structures and Algorithms, Database Management System, Programming for Scientific Computing (Python), Operating System

## EXPERIENCE

**SPEEDLABS** New York, NY
*Real-time market engine for sports, pioneering Momentum Markets* July 2026 - Present
**Quantitative Research Intern**

- Traced a game event across four Python microservices; root-caused an alerting job that had never fired, isolating four defects including a mis-quoted SQL identifier that silently routed 19,000 log rows to a shadow table, then fixed the internal pattern document that prescribed it
- Built the platform's question-quality measurement system, which now runs after every live game and re-grades each settled question from raw play data instead of reusing the settlement code, so it cannot inherit the same bugs
- Disproved a suspected correlation across 202 question pairs and 17 games that leadership was scoping a live safeguard around; traced the real cause to a pricing bias making a 20-question winning streak 110x more likely than a coin-flip model implies, redirecting the fix to pricing

**DEEP MARKET MAKING Inc.** New York, NY
*AI Startup building the world's first Large Event Model for credit market pricing* May 2026 - June 2026
**Quantitative AI Architect Intern**

- Co-developed a macro conditional pricing pipeline to forecast corporate bond shifts across 1-day, 1-week, and 1-month horizons to help revalue illiquid assets
- Built a robust data ingestion engine for core macro factors (SOFR, ICE BofA spreads, VIX, S&P 500) using localized fallbacks to bypass corporate proxy firewall blocks
- Isolated absolute factor levels from conditioning deltas, applying log returns for equities and point changes for volatility to keep features consistent across market regimes

## INDEPENDENT PROJECTS

### IMPLIED VOLATILITY SURFACE AND ARBITRAGE DETECTION

- Constructed a real-time implied volatility surface using live option chain data from Yahoo Finance, computing call option implied volatilities with the Black-Scholes model and Brent's root-finding method
- Modeled volatility as a function of strike price and time to maturity, interpolating a smooth 3D surface using SciPy grid interpolation and visualizing it with Plotly
- Detected butterfly arbitrage by testing convexity conditions across strikes for fixed maturities, identifying violations in the implied volatility smile
- Developed an interactive Streamlit dashboard with dynamic ticker selection, expiry filters, and real-time U.S. Treasury risk-free rates

### MARKET RISK ANALYSIS FOR A FIXED INCOME PORTFOLIO WITH VaR, STRESS TESTING AND SCENARIO ANALYSIS (PYTHON)

- Designed and implemented a Monte Carlo VaR & Expected Shortfall (CVaR) model for a $5M corporate bond portfolio (JPMorgan, Mastercard, Ford, Broadcom, Wells Fargo) with investment-grade ratings (BBB- to A+) and maturities from 2 to 7 years
- Extracted Treasury yield data from FRED API, computed G-Spread and Z-Spread using yield curve interpolation
- Conducted Monte Carlo simulations (10,000 paths) using Geometric Brownian Motion (GBM) to compute 1-Day and 10-Day VaR (99%, 97%, 95% confidence levels), along with Expected Shortfall (CVaR) for tail risk assessment
- Performed stress testing & scenario analysis on interest rate shocks (Parallel, Steepening, Flattening in Treasury yields) and credit spread widening of 75 to 150 bps on bonds individually, assessing portfolio loss impact under extreme market conditions